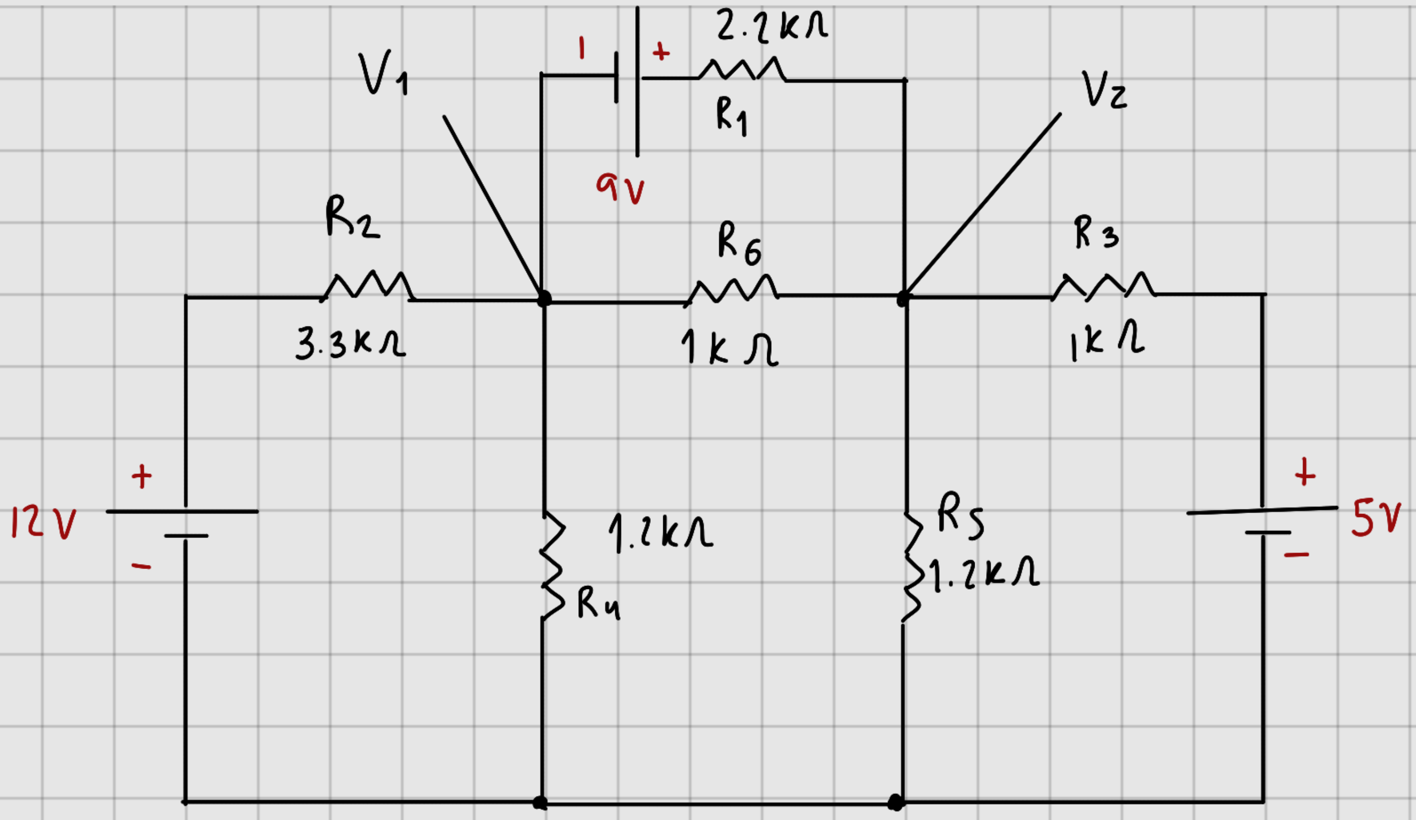
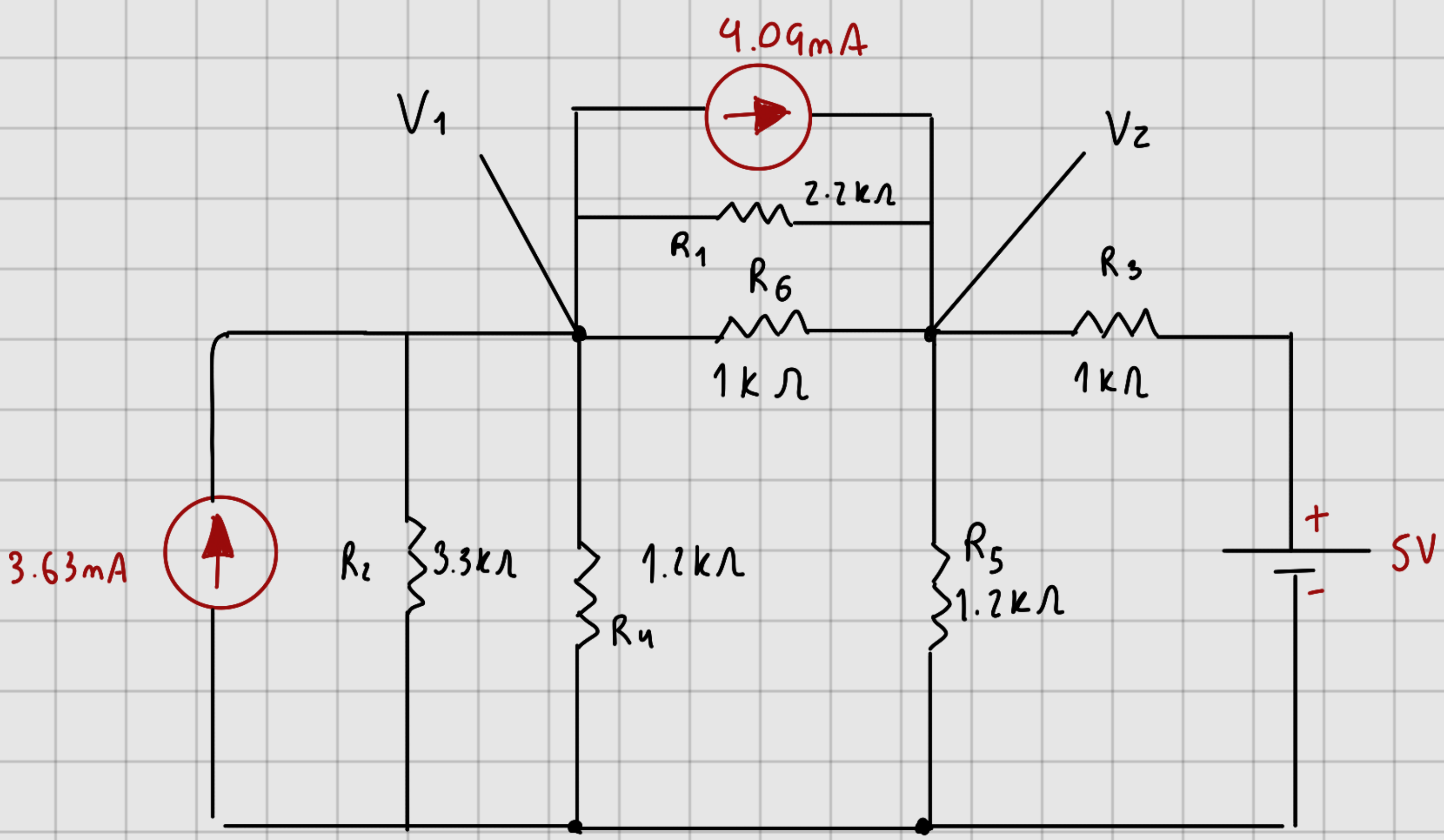
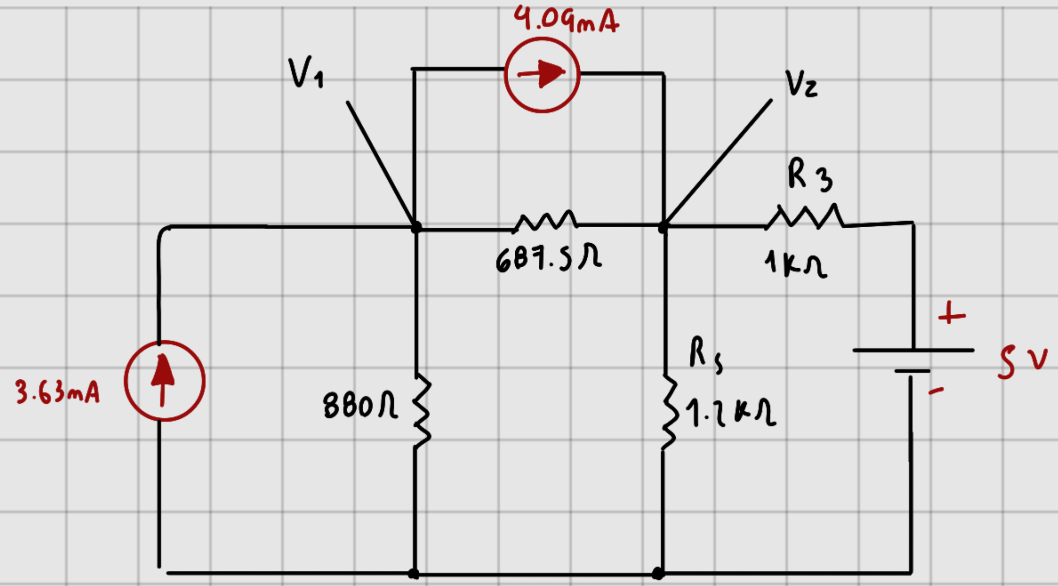


I_{R4}, I_{R5}, I_{R6}, P_{R1}, P_{R2}, P_{R3}, V_1, V_2



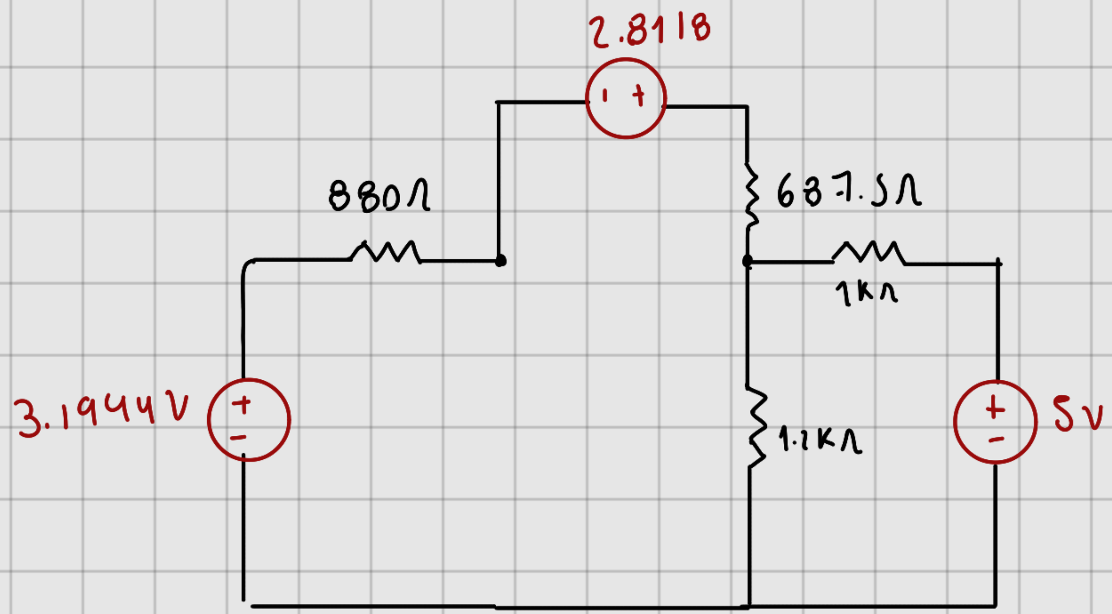
$$I_{S_1} = \frac{12V}{3.3k\Omega} = 3.63mA$$

$$I_{S_3} = \frac{9V}{2.2k\Omega} = 4.09mA$$



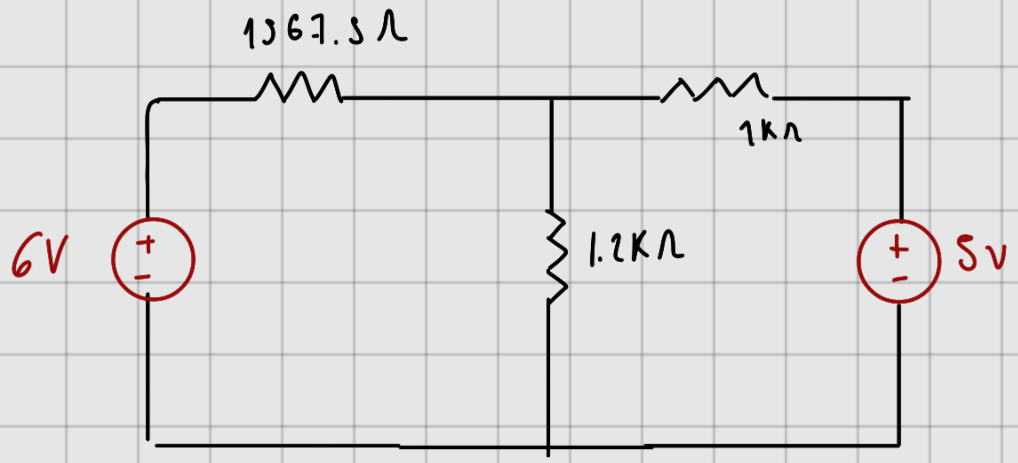
$$R_A = \frac{1}{\frac{1}{3.3k\Omega} + \frac{1}{1.2k\Omega}} = 880\Omega \qquad R_B = \frac{1}{\frac{1}{2.2k\Omega} + \frac{1}{1k\Omega}} = 687.5\Omega$$

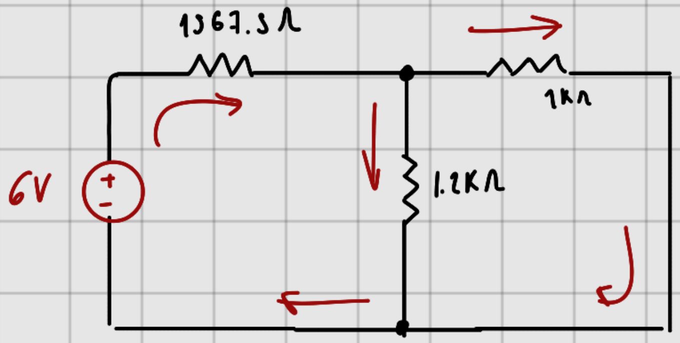
$$R_C = \frac{1}{\frac{1}{1.2k\Omega} + \frac{1}{1k\Omega}} = 545.45\Omega$$

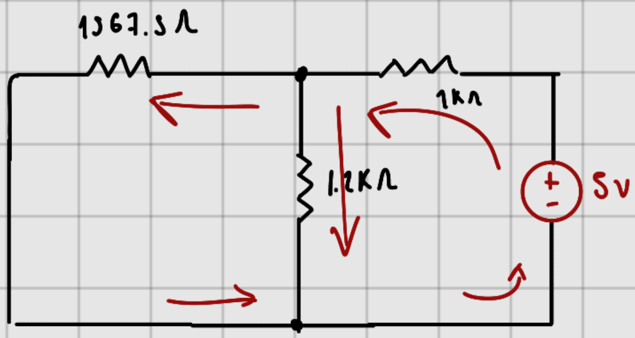


$$V_{s_1} = 3.63mA(880) = 3.1944\ V$$

$$V_{s_2} = 4.09mA(687.5) = 2.8118\ V$$







$$R_T = 1567.5 + \frac{1}{\frac{1}{1.2k\Omega} + \frac{1}{1k\Omega}}$$

$$R_T = 2113\ \Omega$$

$$I_T = \frac{6V}{2113\Omega} = 2.83mA$$

$$V = 1567.5(2.83mA)$$

$$V_2 = 1.56$$

$$R_T = 1k\Omega + \frac{1}{\frac{1}{1567.5\Omega} + \frac{1}{1.2k\Omega}}$$

$$R_T = 1679.67\ \Omega$$

$$I_T = \frac{5V}{1679.67} = 2.97mA$$

$$V_{1k} = 1\cancel{k}\Omega(2.97\cancel{m}A)$$

$$V_{1k} = 2.97\ V$$

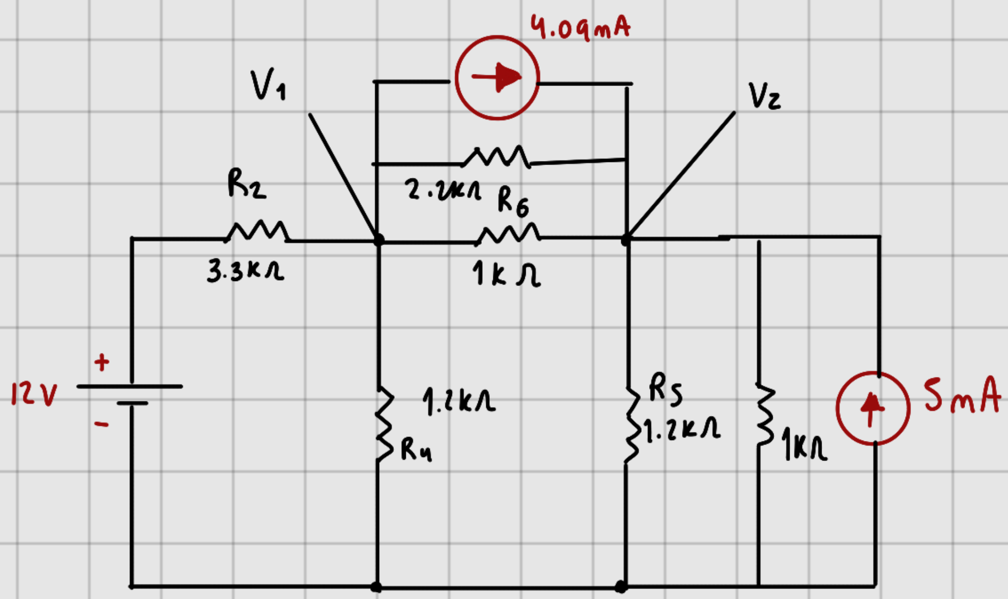
$$V_2 = 2.03$$

$$V_2 = 2.03 + 1.56V = 3.6V$$

$$I_{R_5} = \frac{1.56}{1.2k\Omega} = 1.3mA \qquad I_{R_5} = 1.6mA$$

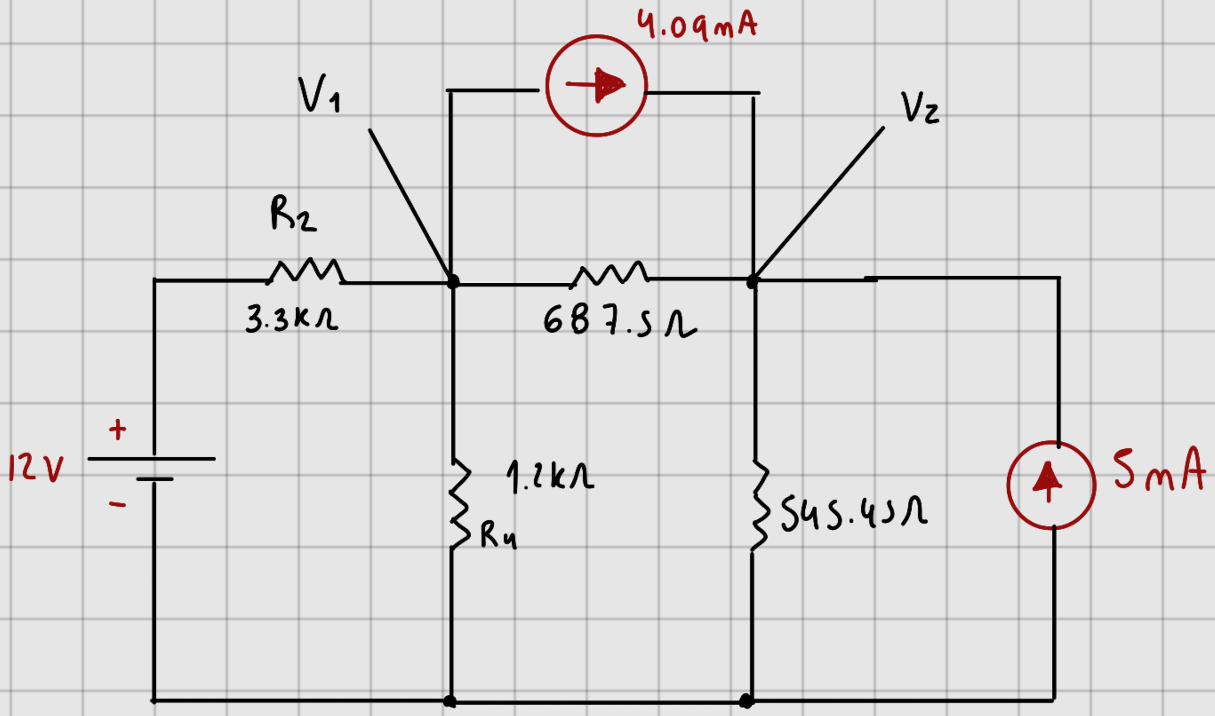
$$I_{R_5} = 1.3mA + 1.6mA = 2.9mA$$

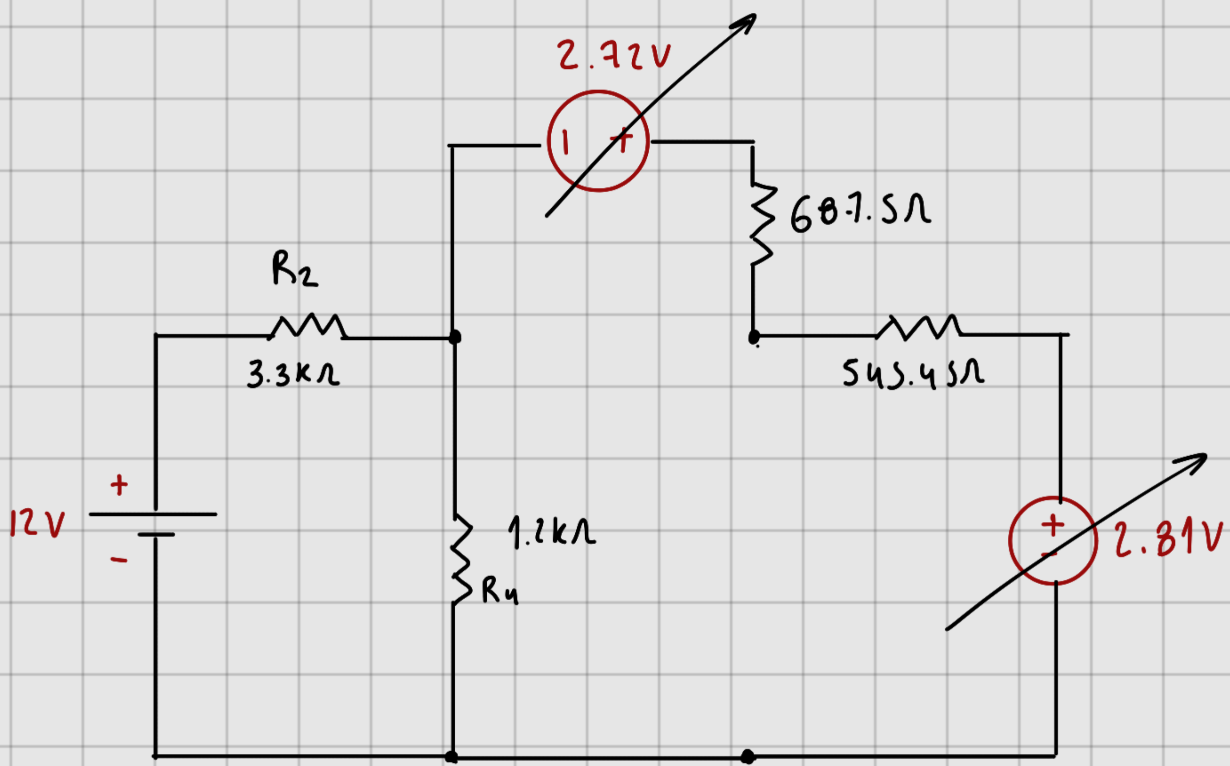
$$P_{R_5} = V_2 I_{R_5} = 10.44mW$$



$$I_s = \frac{V_s}{R} = \frac{5V}{1k\Omega} = 5mA$$

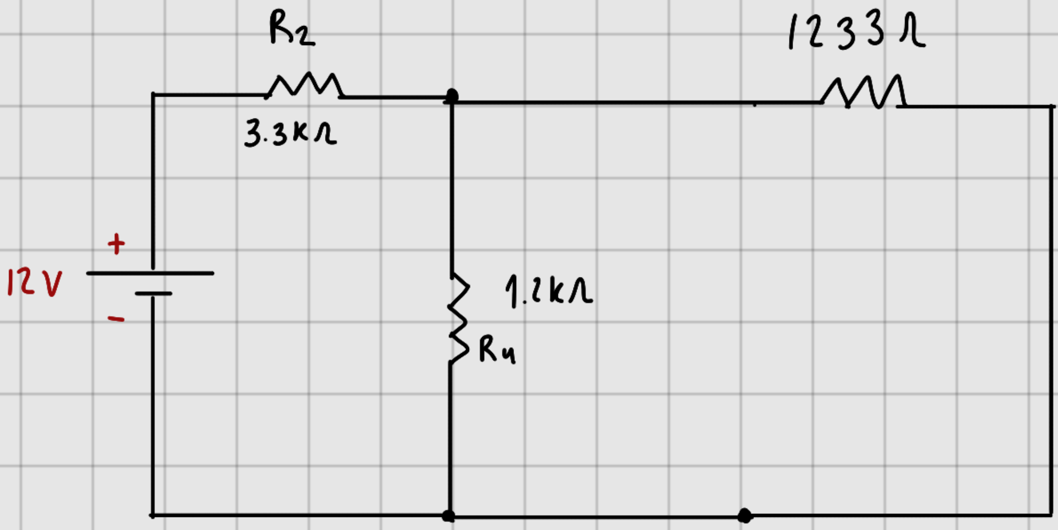
$$R_A = \frac{1}{\frac{1}{2.2k} + \frac{1}{1k\Omega}} = 687.5\Omega \qquad R_B = \frac{1}{\frac{1}{1.2k\Omega} + \frac{1}{1k\Omega}} = 545.45\Omega$$





$$V_{s_1} = 4.09mA(687.5\Omega) = 2.81\ V$$

$$V_{s_2} = 5mA(545.45\Omega) = 2.72\ V$$



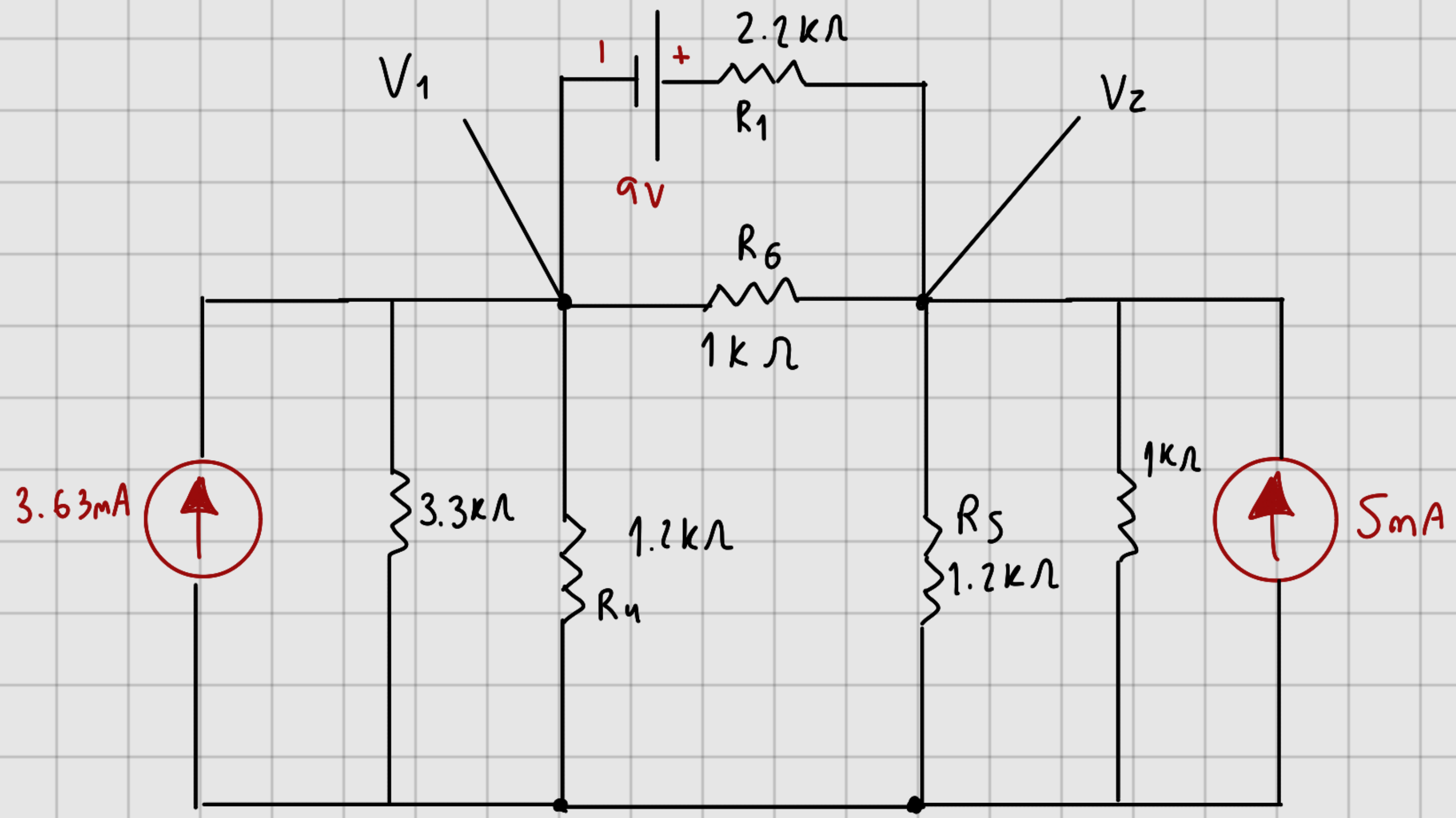
$$R_T = 3.3k\Omega + \frac{1}{\frac{1}{1233\Omega} + \frac{1}{1200\Omega}} = 3908.1\Omega$$

$$I_T = \frac{12V}{3908.1\Omega} = 3.07mA$$

$$V_{R_2} = 10.131V \qquad V_1 = 1.86V$$

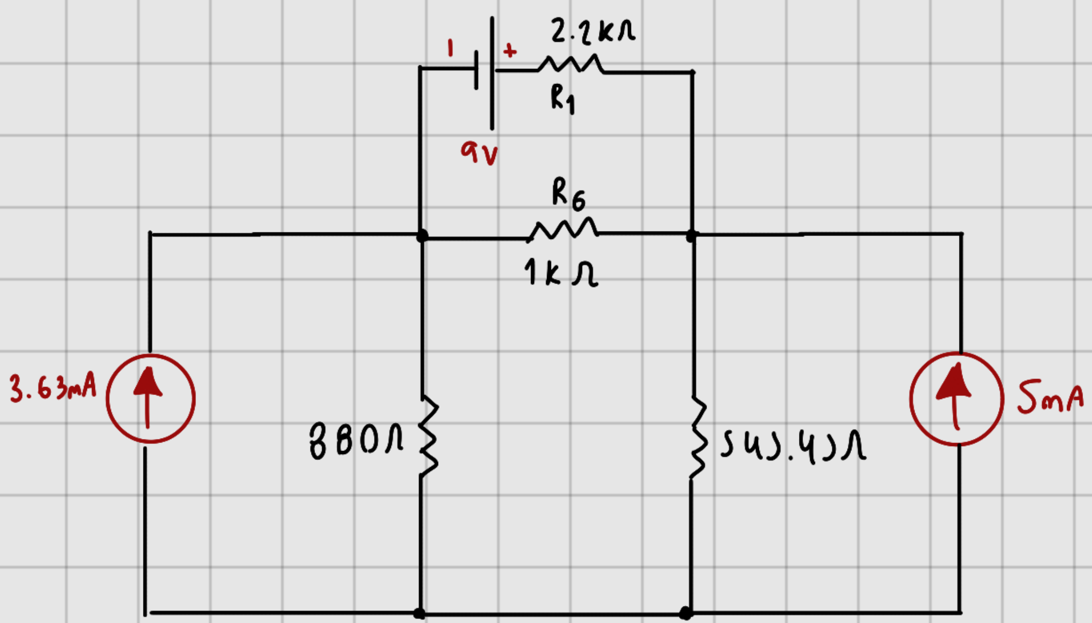
$$I_{R_4} = \frac{1.86}{1200} = 1.5mA$$

$$P_{R_4} = 1.5mA(1.86V) = 2.89mW$$



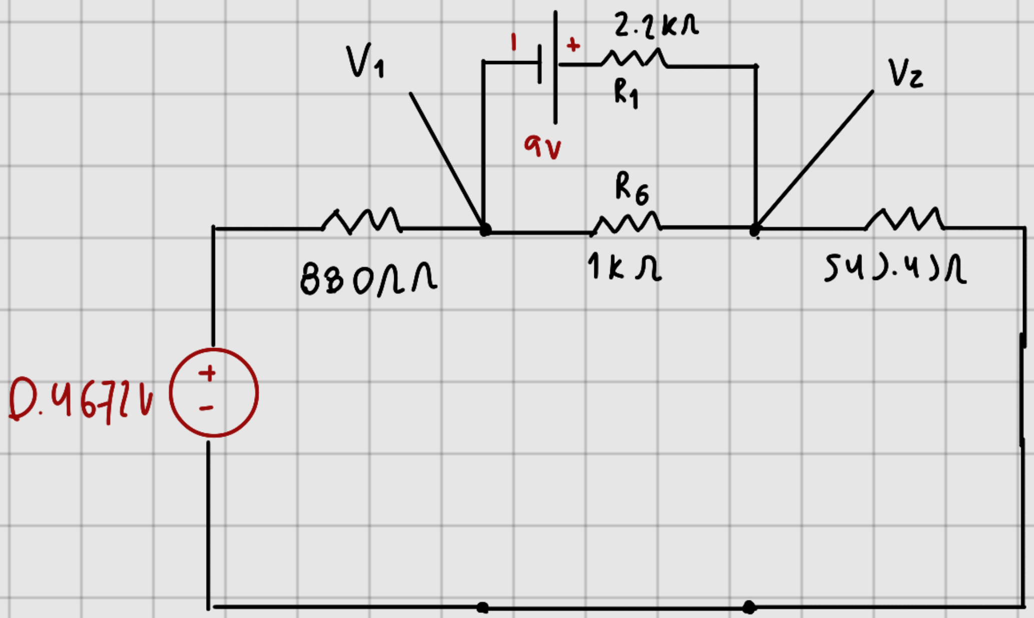
$$R_A = \frac{1}{\frac{1}{3.3k\Omega} + \frac{1}{1.2k\Omega}} = 880\Omega$$

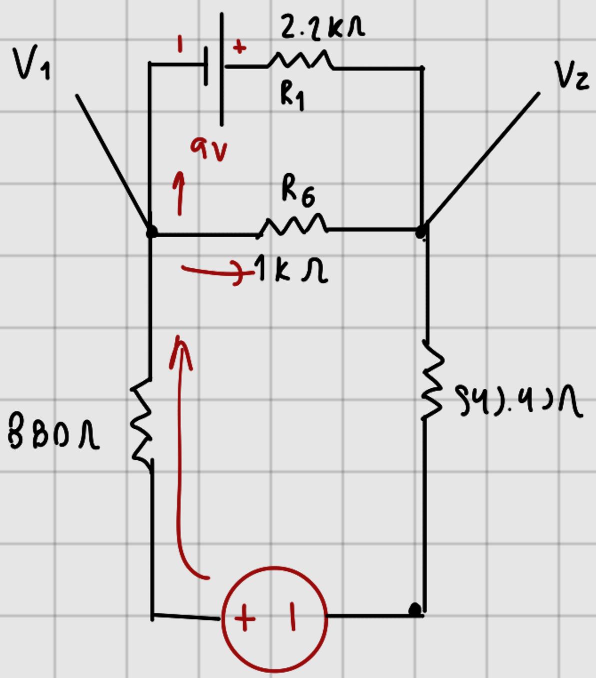
$$R_B = \frac{1}{\frac{1}{1.2k\Omega} + \frac{1}{1k\Omega}} = 545.45\Omega$$

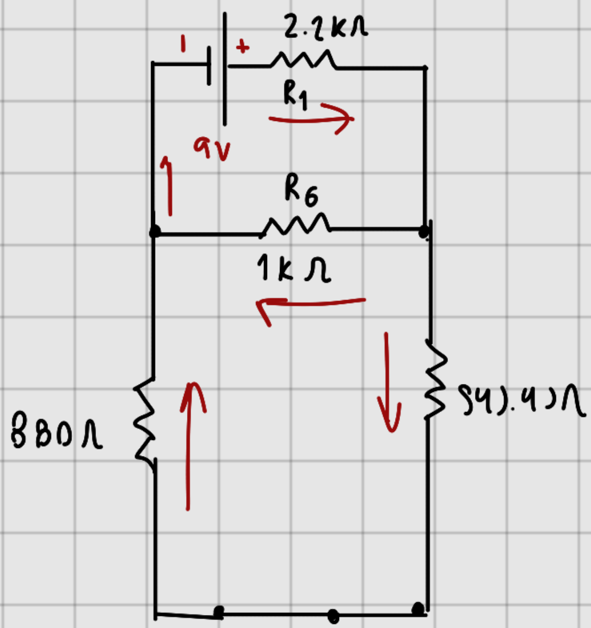


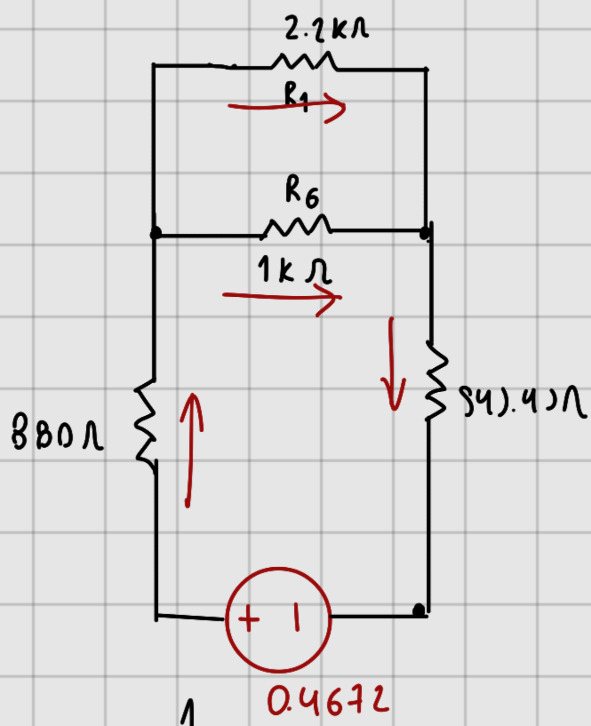
$V_s = 3.63\,mA\,(880\,\Omega)$ $V_s = 5\,mA\,(545.45\,\Omega)$

$V_s = 3.1944\ V$ $V_s = 2.7272\ V$









$R_A = 1425.45$

$R_T = 2787.7\,\Omega$

$I_T = \dfrac{9V}{2787.7} = 3.22mA$

$V_{R1} = 3.22\cancel{m}A(2.2\cancel{k}\Omega)$

$V_{R1} = 7.084\ V$

$V_2 = 1.916\ V$

$I_{R6} = \dfrac{1.916V}{1000\Omega} = 1.9mA$

$R_A = \dfrac{1}{\frac{1}{2.2k\Omega} + \frac{1}{1k\Omega}} = 687.5$

$R_T = 2113\,\Omega$

$I_T = \dfrac{0.4672}{2113\,\Omega} = 0.2mA$

$V_{R1} = 880(0.2mA) = 0.176V$

$V_1 = 0.2912\ V$

$I_{R6} = \dfrac{0.2912V}{1k} = 0.2mA$

$I_{R6} = 1.91mA - 0.2mA = 1.71mA$

$P_{R6} = 1.71mA(1.625) = 2.77mW$

$V_1 = 1.86\,V$

$V_2 = 3.6\,V$

$I_{R4} = 1.5\,mA$

$I_{R5} = 2.9\,mA$

$I_{R6} = 1.71\,mA$

$P_{R4} = 2.89\,mW$

$P_{R5} = 10.44\,mW$

$P_{R6} = 2.77\,mW$